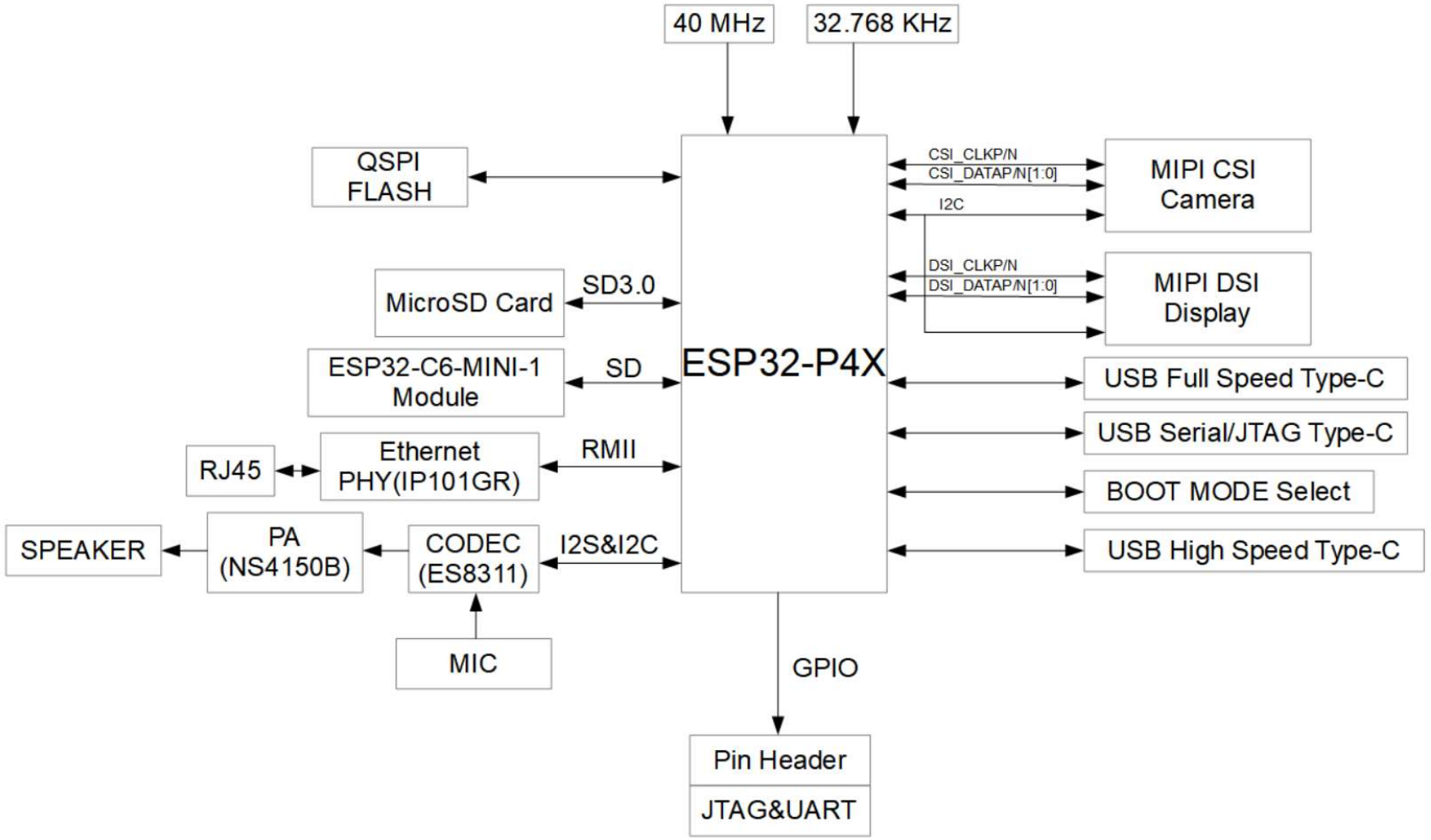
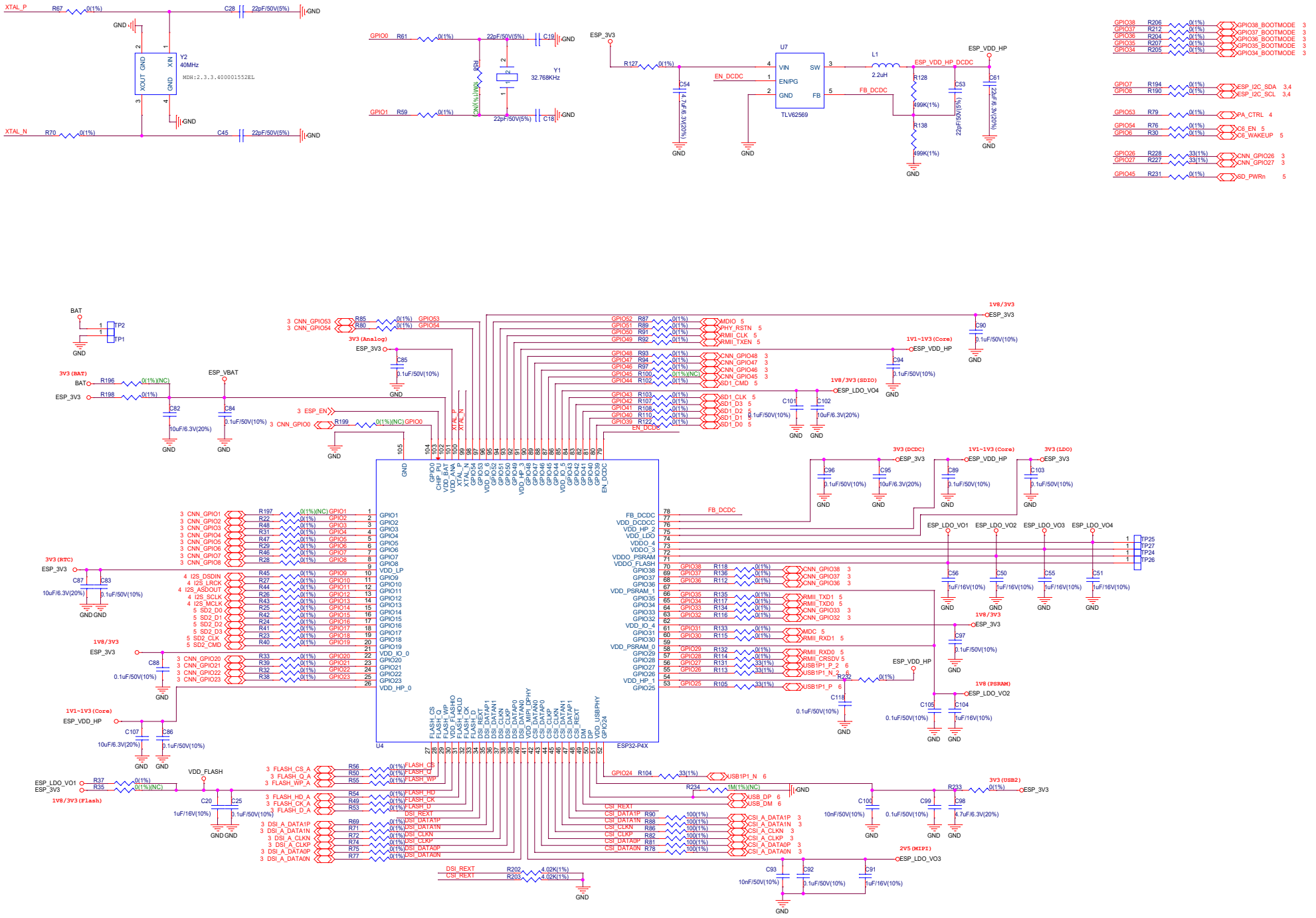
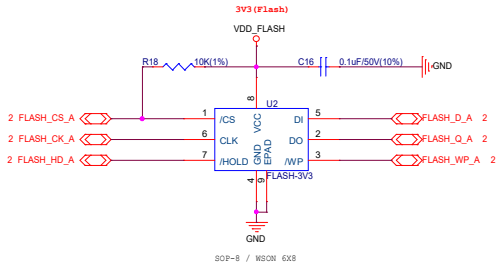
Design Block:



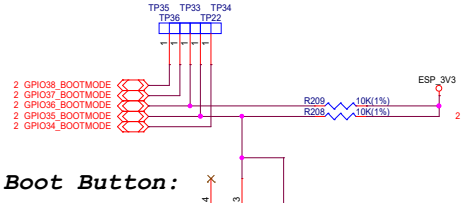
ESPRESSIF			
乐鑫信息科技（上海）股份有限公司			
Title: ESP32-P4X_FUNCTION_EV_BOARD			
Size: C	Page Name: 01_DESIGNBLOCK	Rev: 1.8	Proprietary
Date: Wednesday, August 05, 2026		Sheet 1 of 6	


XTAL_P
XTAL_N
Y2
40MHz
Y1
32.768KHz
U7
TLV62569
VIN
SW
EN/PG
GND
FB
L1
2.2uH
ESP_VDD_HP
ESP_VDD_HP_DCDC
FB_DCDC
EN_DCDC
ESP_3V3
BAT
U4
ESP32-P4X
FLASH_CS
FLASH_Q
FLASH_WP
FLASH_HOLD
FLASH_CK
FLASH_D
DSI_REXT
CSI_REXT
VDD_FLASH
ESP_VBAT
ESP_LDO_VO1
ESP_LDO_VO2
ESP_LDO_VO3
ESP_LDO_VO4
ESPRESSIF
乐鑫信息科技（上海）股份有限公司
ESP32-P4X_FUNCTION_EV_BOARD
Wednesday, August 05, 2026

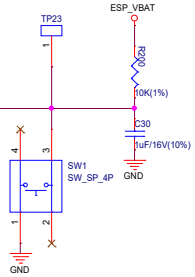
FLASH:



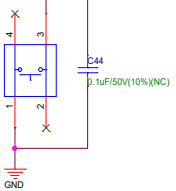
Boot Strapping Pins:



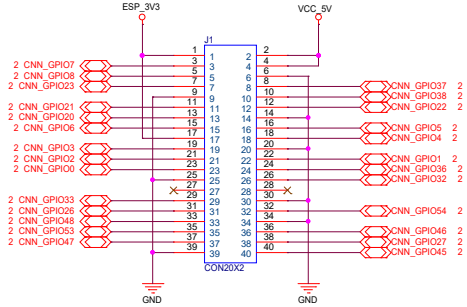
Reset Button:



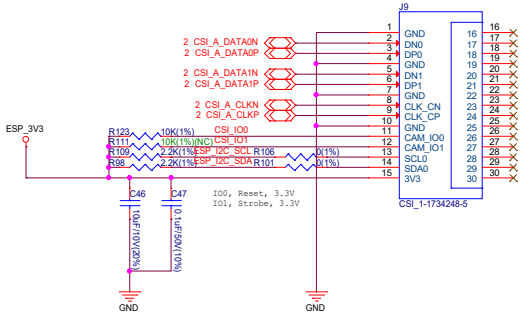
Boot Button:



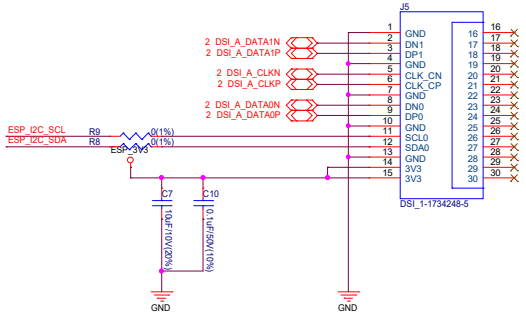
Pin Header For GPIOs:



CSI Connector:



DSI Connector:



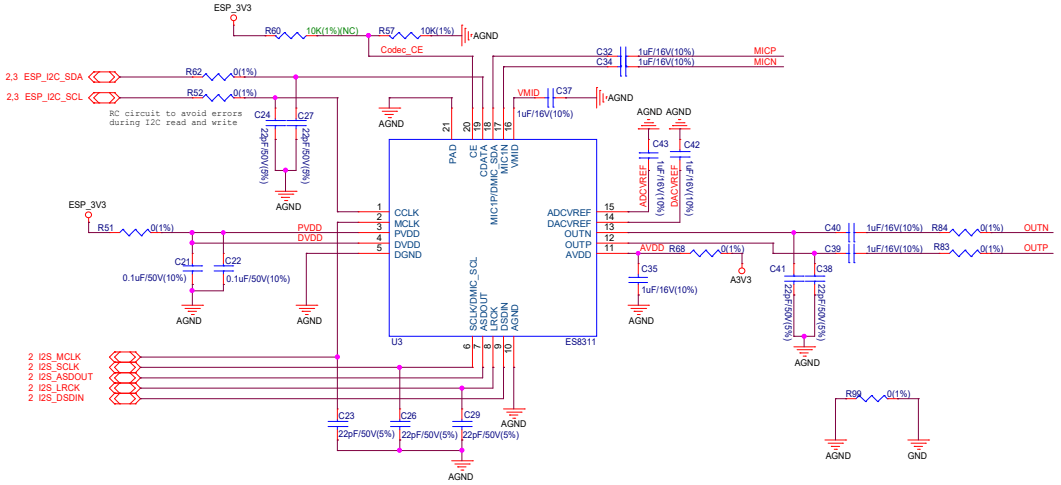


ESPRESSIF

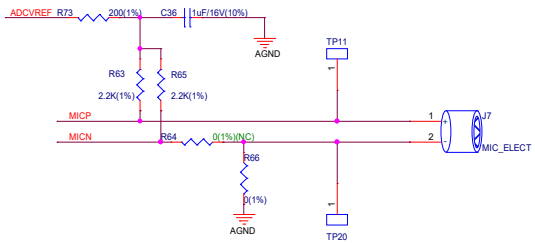
乐鑫信息科技（上海）股份有限公司

Title	ESP32-P4X_FUNCTION_EV_BOARD		
Size: C	Page Name: 03_FLASH_DBG_CNN	Rev: 1.8	Proprietary
Date: Wednesday, August 05, 2026		Sheet 3 of 6	

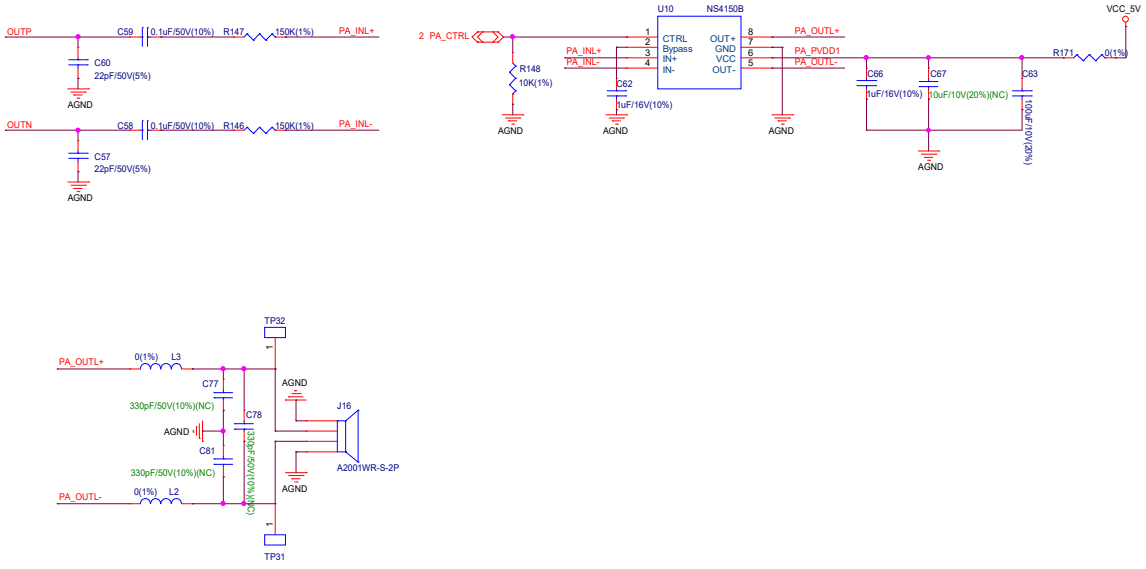
Codec:



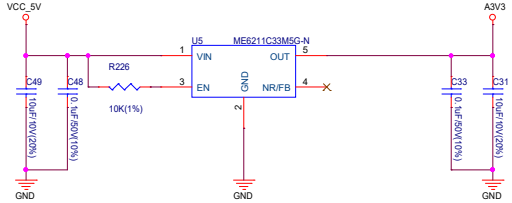
MIC:



PA&SPEAKER:

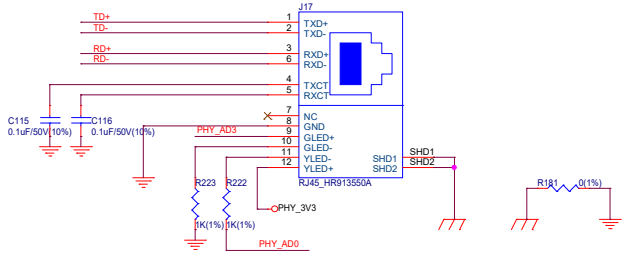


3.3V LDO For AVDD:

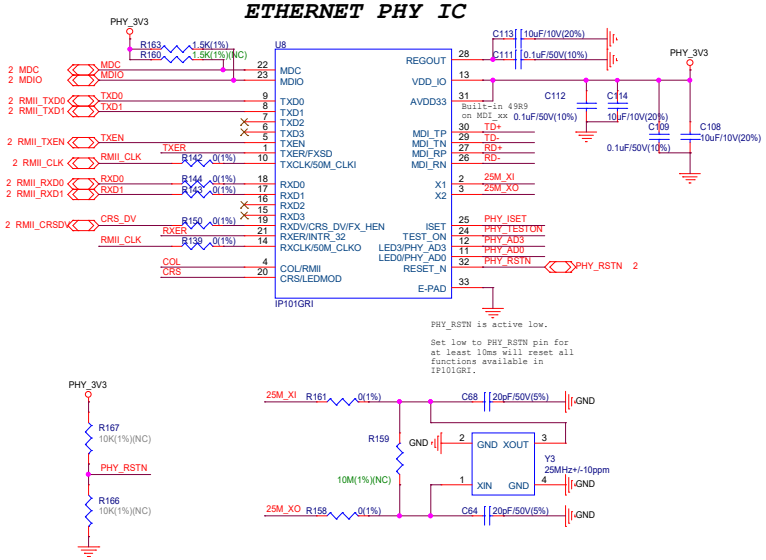


ESPRESSIF			
乐鑫信息科技（上海）股份有限公司			
Title: ESP32-P4X_FUNCTION_EV_BOARD			
Size: C	Page Name: 04_Audio	Rev: 1.8	Proprietary
Date: Wednesday, August 05, 2026		Sheet 4 of 6	

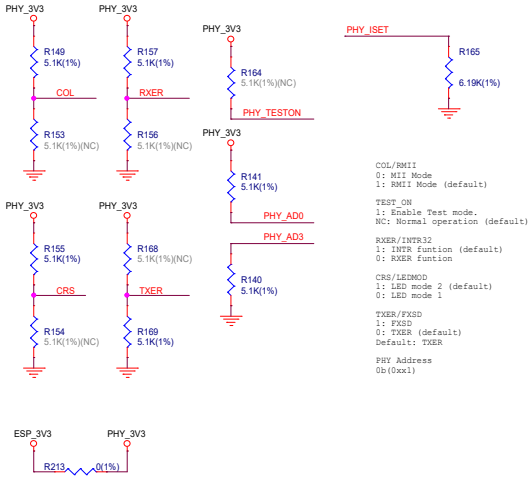
RJ45 with Trans



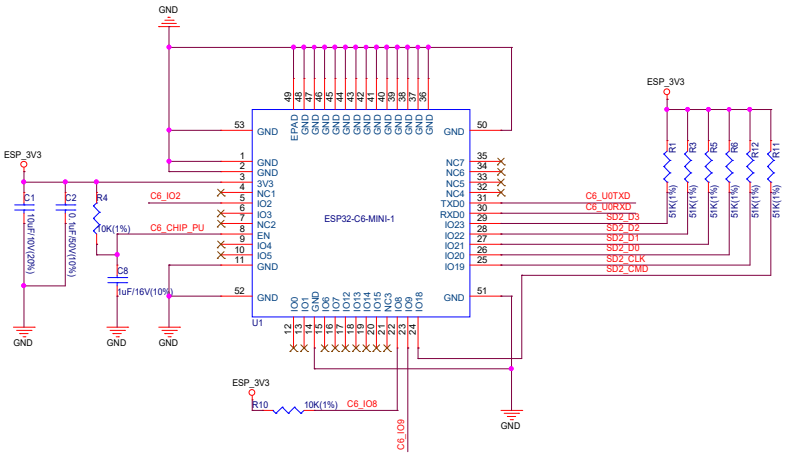
ETHERNET PHY IC

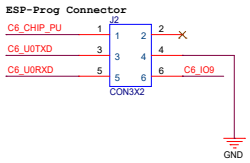


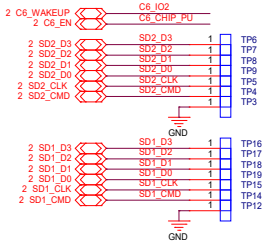
PHY MODE CONFIG



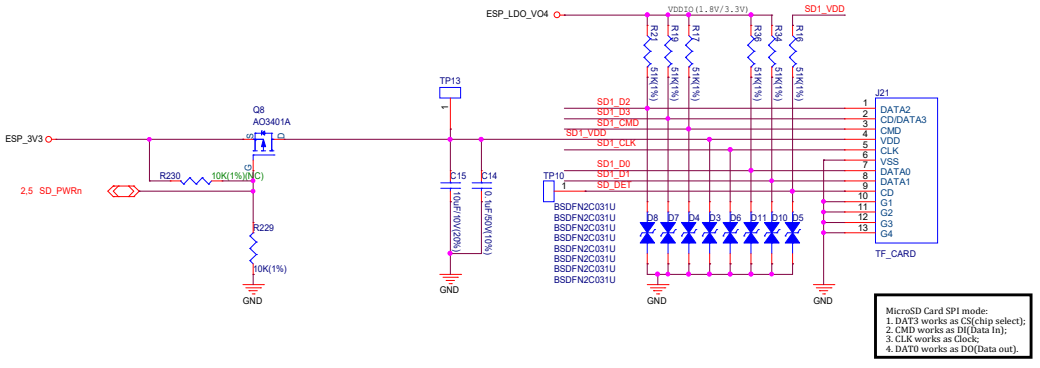
WiFi & BT Module:







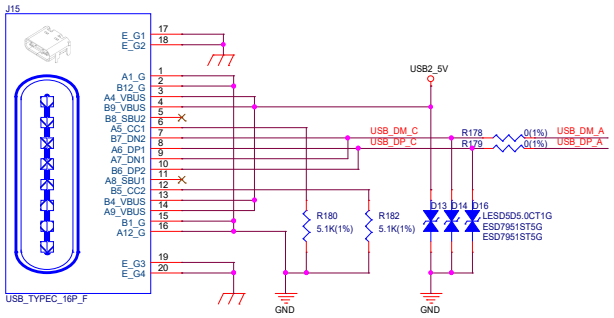
MicroSD Card:



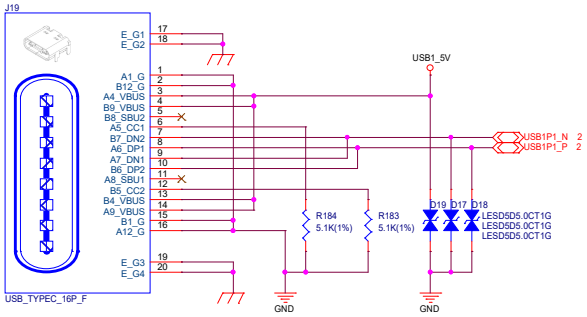
ESPRESSIF
乐鑫信息科技（上海）股份有限公司

Title	ESP32-P4X_FUNCTION_EV_BOARD		
Size	Page Name	Rev	Proprietary
C	05_Ethernet_SDMMC_WiFi	1.8	
Date:	Wednesday, August 05, 2026	Sheet	5 of 6

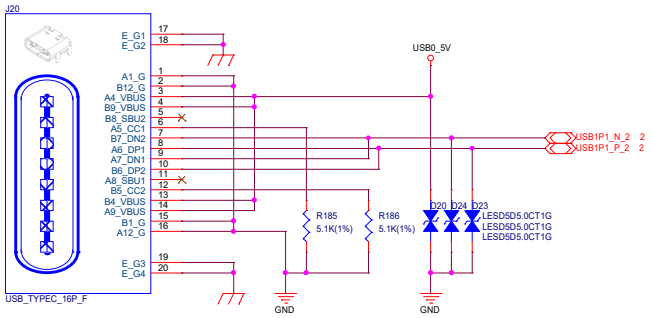
USB High Speed & Power in:



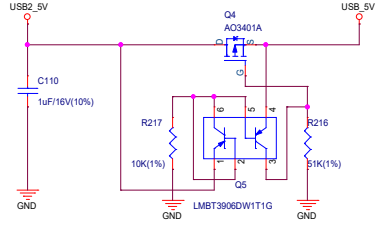
USB Serial/JTAG & Power in:

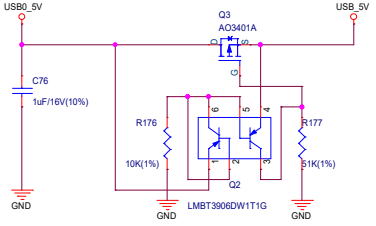


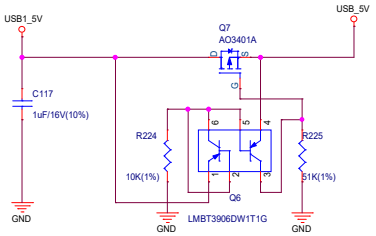
USB Full Speed Slave & Power in:

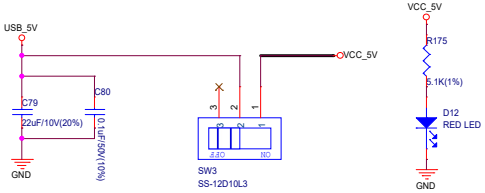


Ideal Diodes:

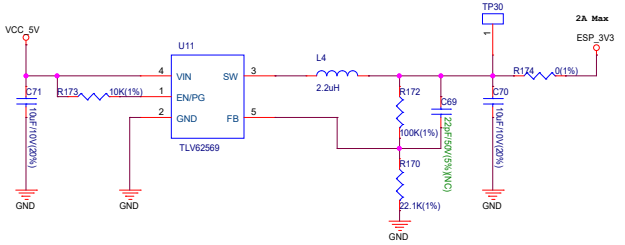




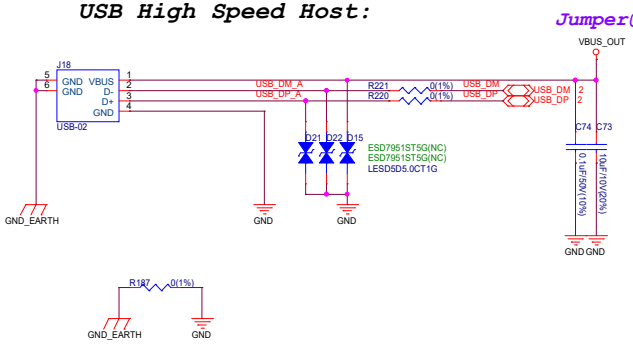




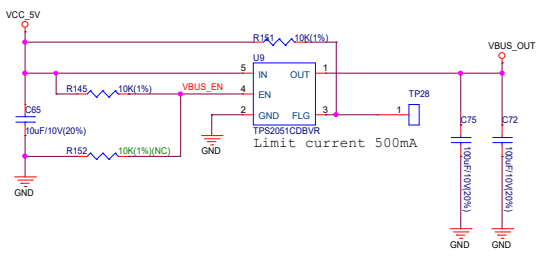
3V3 DC-DC:

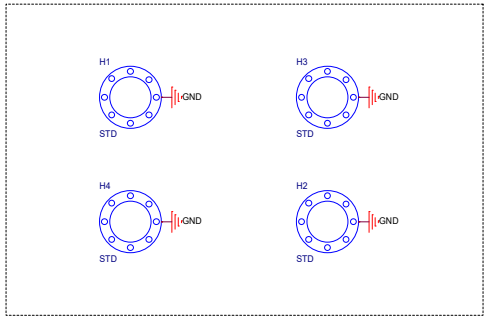


USB High Speed Host:



Jumper@J10





ESPRESSIF
乐鑫信息科技（上海）股份有限公司

Title	ESP32-P4X_FUNCTION_EV_BOARD		
Size C	Page Name 06_SYS_PWR_USB	Rev 1.8	Proprietary
Date: Wednesday, August 05, 2026		Sheet 6 of 6	